The bicycle plant

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The bicycle plant is the controllers' own dynamic single-track model with three effects added: actuator saturation, a 50 ms steering lag and tire force build-up over 0.3 m. It is a 10-state ODE stepped by fourth-order Runge–Kutta at 1 ms, and a 50 ms world step costs about 0.03 ms, roughly 1660× real time.

Summary

- **Where it sits.** `sim::BicyclePlant` (`sim/plant/bicycle_plant.cpp`) implements the same `PlantModel` interface as the Chrono plant (`sim/plant/interface.hpp`): `reset`, `observe`, `saturate`, `step`, `set_friction`.
- **Who uses it.** `build/simulate` by default, and `--dynamics bicycle` in CARLA (`BicycleDynamics` in `carla_bridge/world.py`), where CARLA only draws the car flat on the road.
- **Input.** u = [δ̇, Fx]: the road-wheel steering rate [rad/s] and the total longitudinal tire force [N], positive to drive, negative to brake.
- **Observed state.** x = [X, Y, ψ, Ux, Uy, r, δ]: position of the centre of gravity, heading, body-frame velocities, yaw rate and road-wheel angle.
- **Hidden state.** Three more numbers the controller never sees: the commanded steering angle δref and the two lagged slip angles α̃f, α̃r.
- **Same equations as the NMPC.** The vehicle equations are the templates in `track_control/model/bicycle.hpp`, evaluated in double here and in CasADi symbols inside the NMPC. The plant differs from the controller's model only by the lags and limits below, and by any parameters it is given that differ from the controller's.

Geometry and state

The model replaces each axle's two tires with one tire on the centreline, so the car is a rigid body moving in the plane with three degrees of freedom: X, Y and ψ.

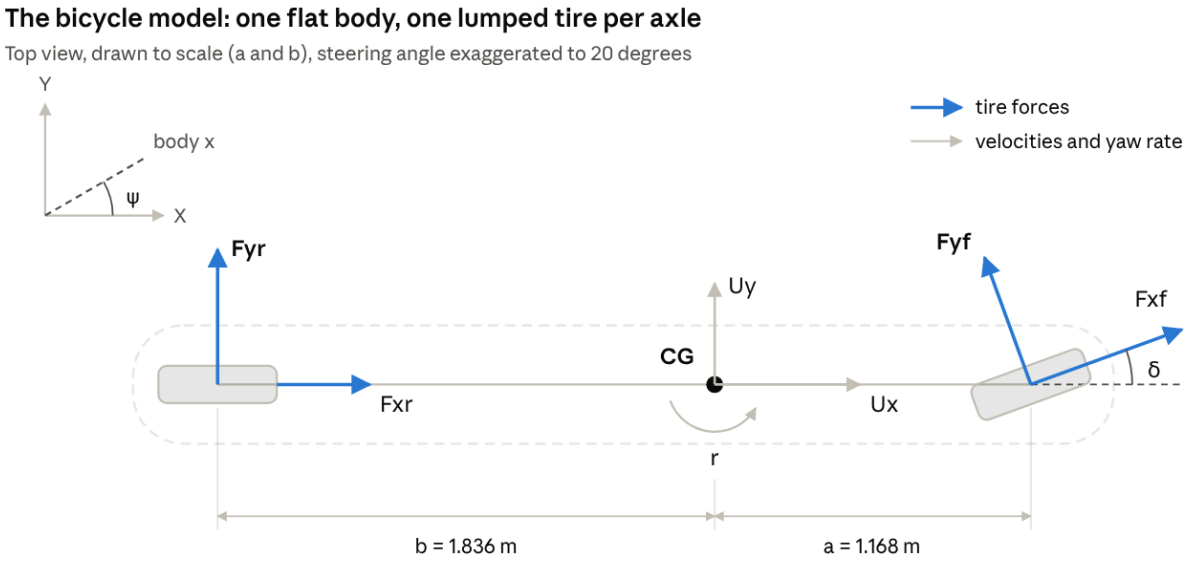


Figure 1. Single-track geometry. Model 3 dimensions from `config/vehicles/model3.json`.

Forces act at the two axles. Velocities are measured at the centre of gravity, in the body frame.

Symbol	Meaning	Unit
X, Y	position of the CG in the world frame	m
ψ	heading: body x-axis from world X, counter-clockwise	rad
Ux, Uy	forward and lateral velocity of the CG, body frame	m/s
r	yaw rate, ψ̇	rad/s
δ	road-wheel steering angle of the front axle	rad
αf, αr	slip angle: angle between where an axle's tire points and where it moves	rad
Fxf, Fxr	longitudinal tire force on each axle, along the wheel	N
Fyf, Fyr	lateral tire force on each axle, across the wheel	N
Fzf, Fzr	normal load on each axle	N
a, b, L	CG to front axle, CG to rear axle, wheelbase a + b	m

Equations of motion

Every derivative is an explicit formula of the state and the input, so stepping the plant never solves an equation. From `dynamics()` in `track_control/model/bicycle.hpp`:

Splitting Fx between the axles. A fixed share goes to the front axle, `drive_front` when driving and `brake_front` when braking (both 0.5 for the Model 3):

$$F_{xf} = s\,F_x, \qquad F_{xr} = F_x - F_{xf}, \qquad s = \begin{cases} \text{drive_front} & F_x \geq 0 \\ \text{brake_front} & F_x < 0 \end{cases}$$

Normal loads, with quasi-static longitudinal load transfer. Braking moves load to the front axle; driving moves it to the rear:

$$F_{zf} = \frac{mg\,b - h\,F_x}{L}, \qquad F_{zr} = \frac{mg\,a + h\,F_x}{L}, \qquad L = a + b$$

Slip angles. Ux is floored at Ux,min = 1 m/s so the model stays defined at standstill:

$$\alpha_f = \arctan\frac{U_y + a\,r}{U_x} - \delta, \qquad \alpha_r = \arctan\frac{U_y - b\,r}{U_x}$$

Resistance. Aerodynamic drag plus rolling resistance, acting along the body x-axis:

$$F_{\text{drag}} = \tfrac{1}{2}\,\rho\,C_d A\,U_x^2 + C_{rr}\,mg$$

Body-frame dynamics. The front axle's forces are rotated by the steering angle. The rear axle is not steered:

$$\begin{aligned} F_{yf}^{\text{body}} &= F_{yf}\cos\delta + F_{xf}\sin\delta \\ \dot{U}_x &= \frac{F_{xf}\cos\delta - F_{yf}\sin\delta + F_{xr} - F_{\text{drag}}}{m} + r\,U_y \\ \dot{U}_y &= \frac{F_{yf}^{\text{body}} + F_{yr}}{m} - r\,U_x \\ \dot{r} &= \frac{a\,F_{yf}^{\text{body}} - b\,F_{yr}}{I_{zz}} \end{aligned}$$

World-frame kinematics. The body velocity is rotated by the heading. The steering angle integrates the commanded rate, after the actuator lag below:

$$\dot{X} = U_x\cos\psi - U_y\sin\psi, \qquad \dot{Y} = U_x\sin\psi + U_y\cos\psi, \qquad \dot{\psi} = r, \qquad \dot{\delta} = u_{\dot{\delta}}$$

The terms r·Uy and −r·Ux are the centripetal terms that appear because the velocities are written in a rotating frame. There is no roll, pitch or heave: the body stays flat, and each axle is a single lumped tire.

Tire model

Each axle's lateral force comes from the Fiala brush model (`track_control/model/tire.hpp`). It is linear in the slip angle for small slip and flat at the friction limit once the whole contact patch slides.

Friction-circle derating. The longitudinal force on an axle uses up part of its grip. What is left for cornering is the rest of the friction circle, smoothed near zero so the model stays differentiable for the NMPC (F_min = 50 N):

$$F_{y,\max} = \sqrt{\tfrac{1}{2}\left(s + \sqrt{s^2 + 4F_{\min}^4}\right)}, \qquad s = (\mu F_z)^2 - F_x^2 \quad \left(\approx \sqrt{(\mu F_z)^2 - F_x^2}\right)$$

Fiala lateral force. With z = tan α and the slip at which the patch slides completely, z_sl = 3 F_y,max / Cα:

$$F_y = \begin{cases} -C_\alpha z + \dfrac{C_\alpha^2}{3F_{y,\max}}\,|z|\,z - \dfrac{C_\alpha^3}{27F_{y,\max}^2}\,z^3 & |z| < z_{sl} \\ -F_{y,\max}\,\operatorname{sign}(z) & |z| \geq z_{sl} \end{cases}$$

The force opposes the slip, hence the minus sign. In the plant the slip angles fed to this formula are the lagged ones, α̃f and α̃r.

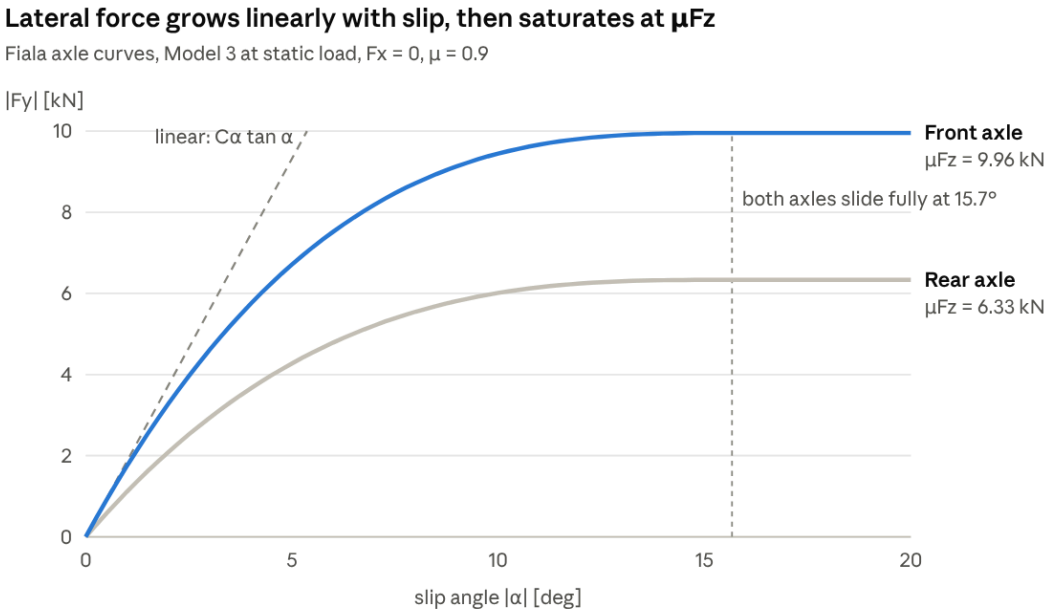


Figure 2. Computed from the Fiala formula with `config/vehicles/model3.json`: static loads, no drive or brake force.

Braking or driving shrinks F_y,max and moves the load between axles, so the same curves flatten earlier under braking at the rear and under drive at the front. The model has no longitudinal slip: Fx is applied as commanded, within the limits of the next section.

Actuators, lags and limits

The plant adds three things the NMPC's model leaves out (`BicyclePlant::deriv` and `saturate`). Each can be switched off by giving it 0.

1. Input saturation, applied to u before every 1 ms step, using the current speed:

$$u_{\dot\delta} \in [-\dot\delta_{\max},\ \dot\delta_{\max}], \qquad F_x \in [-F_{\text{brake}},\ F_{\text{drive}}(U_x)]$$

The force limits come from `fx_limits()` (`track_control/src/bicycle.cpp`). Each is the smallest of what either axle can pass to the road at its load (fixed split, load transfer included), the actuator peak, and, for drive, the power limit:

$$F_{\text{brake}} = \min\Big(\frac{\mu m g\, b}{f_b L - \mu h},\ \frac{\mu m g\, a}{(1-f_b)L + \mu h},\ F_{\text{brake,max}}\Big)$$

$$F_{\text{drive}}(U_x) = \min\Big(\frac{\mu m g\, b}{f_d L + \mu h},\ \frac{\mu m g\, a}{(1-f_d)L - \mu h},\ F_{\text{drive,max}},\ \frac{P_{\max}}{\max(U_x,\ U_{x,\min})}\Big)$$

For the Model 3 on asphalt (μ = 0.9):

Limit	Value	Set by
Brake force	7568 N (0.42 g)	F_brake,max; the rear axle's grip would allow 9977 N
Drive force below 44.6 m/s	15 683 N (0.87 g)	front axle's grip
Drive force above 44.6 m/s	700 kW / Ux	power
Steering rate	±0.6 rad/s	delta_rate_max
Steering angle	±0.855 rad	delta_max

2. Steering actuator lag. The input rate drives a reference angle, which stops at the lock. The road-wheel angle follows the reference with a first-order lag, τ = 50 ms:

$$\dot{\delta}_{\mathrm{ref}} = \begin{cases} 0 & |\delta_{\mathrm{ref}}| \geq \delta_{\max} \text{ and moving outward} \\ u_\delta & \text{otherwise} \end{cases}, \qquad \dot{\delta} = \frac{\delta_{\mathrm{ref}} - \delta}{\tau_{\mathrm{steer}}}$$

This lagged rate replaces u in the last line of the equations of motion. The NMPC assumes δ̇ = u exactly.

3. Tire relaxation. A tire needs to roll some distance before its lateral force builds up. The tire model sees lagged slip angles that follow the kinematic ones over a relaxation length σ = 0.3 m:

$$\dot{\tilde{\alpha}}_i = \frac{\max(U_x,\, U_{x,\min})}{\sigma}\left(\alpha_i - \tilde{\alpha}_i\right), \qquad i \in \{f, r\}$$

The time constant is σ / Ux: 30 ms at 10 m/s, 6 ms at 50 m/s.

Friction. `set_friction(μ)` changes μ between steps. In CARLA it is set from the surface under the car (asphalt or grass) before every 10 ms call. It scales the tire forces and the force limits.

Integration and stepping

The plant integrates a 10-element internal state with classical fourth-order Runge–Kutta at a fixed dt = 1 ms (`rk4_step` in `track_control/model/integrate.hpp`):

$$z = [\,X,\, Y,\, \psi,\, U_x,\, U_y,\, r,\, \delta,\, \delta_{\mathrm{ref}},\, \tilde{\alpha}_f,\, \tilde{\alpha}_r\,], \qquad z_{k+1} = z_k + \tfrac{dt}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

- `step(u, duration)` runs round(duration / dt) of these steps, holding u. It saturates u again before each one, because the power limit depends on the current speed.
- `observe()` returns the first 7 elements; the controller never sees δref or the lagged slip angles.
- `reset(x0)` sets δref = δ and the lagged slip angles to their kinematic values, so the car starts in equilibrium with its tires.
- **Cost.** Each 1 ms step is 4 evaluations of the formulas above, with no equation solved. A 50 ms world step (50 RK4 steps, 200 evaluations) takes about 0.03 ms, against 61 ms for the Chrono plant (`docs/chrono_plant.pdf`, Table 5).
- **Why 1 ms.** The fastest dynamics are the tire relaxation at high speed (σ / Ux = 6 ms at 50 m/s) and the 50 ms steering lag. RK4 at 1 ms resolves both with a wide margin.

Parameters

The plant reads the same `VehicleParams` as the controllers, from `config/vehicles/model3.json` (CARLA 0.9.15's `vehicle.tesla.model3`; derivations in `docs/vehicle_parameters.md`). The plant's own three settings are constructor defaults in `bicycle_plant.hpp`.

Symbol	Parameter	Value	Source
m	mass	1845 kg	CARLA
Izz	yaw inertia	3958 kg·m²	assumed
a, b	CG to front / rear axle	1.168 m / 1.836 m	derived
h	CG height	0.45 m	assumed
Cαf, Cαr	cornering stiffness per axle	106 500 / 67 800 N/rad	derived
μ	tire–road friction (asphalt)	0.9	assumed
f_d, f_b	front share of drive / brake force	0.5 / 0.5	assumed / derived
F_drive,max	peak drive force	18 073 N	derived
F_brake,max	peak brake force	7568 N	derived
P_max	drive power	700 kW	derived
CdA	drag area	0.483 m²	derived
Crr	rolling resistance	0.012	assumed
ρ	air density	1.225 kg/m³	constant
δmax	road-wheel angle limit	0.855 rad	derived
δ̇max	steering rate limit	0.6 rad/s	design
Ux,min	speed floor in slip angles	1 m/s	design
dt	integration step	1 ms	plant default
τ_steer	steering lag	50 ms	plant default
σ	tire relaxation length	0.3 m	plant default

With these values the static axle loads are 11 062 N front and 7037 N rear, and the car is almost exactly neutral in its linear range: understeer gradient 7.6×10^{-6} rad per m/s².

What it leaves out, compared with the Chrono plant

The bicycle plant is about 2000× cheaper than Chrono because it lumps the car into one flat body with two tires. What that costs:

Effect	Bicycle plant	Chrono plant
Bodies and degrees of freedom	1 body, 3 planar DOF	20 bodies, 13 DOF (126 coordinates, 113 constraint rows)
Roll, pitch, heave	none: the body stays flat	yes
Lateral load transfer	none: left and right wheels lumped per axle	yes, per wheel
Longitudinal load transfer	quasi-static, instant	through the suspension, with its dynamics
Tires	Fiala, lateral only, friction-circle derated	Pacejka 2002 per wheel: combined slip, camber, aligning moment
Longitudinal tire slip	none: Fx is applied directly	yes: wheel spin and lock
Suspension kinematics (camber, toe, roll steer)	none	yes
Differential	none: Fx split by a fixed share	open differential through the driveline
Steering	first-order lag on the angle	rack and linkage, compliant under load
Solver	explicit RK4, no equation solved	one 239-unknown solve per 1 ms step
Cost per 50 ms world step	about 0.03 ms	about 61 ms

Because the plant and the NMPC share the vehicle equations, a lap on the bicycle plant tests the controller against its own model plus the lags and limits. The Chrono plant tests it against a car it does not model.